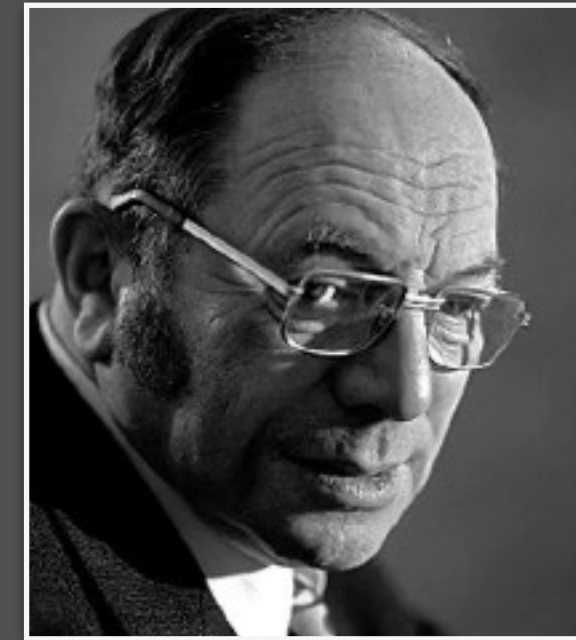


Brenier's Theorem

Assumptions:

1. $c(x, y) = \|x - y\|^2$
2. α has a density $\rho_\alpha = \frac{d\alpha}{dx}$



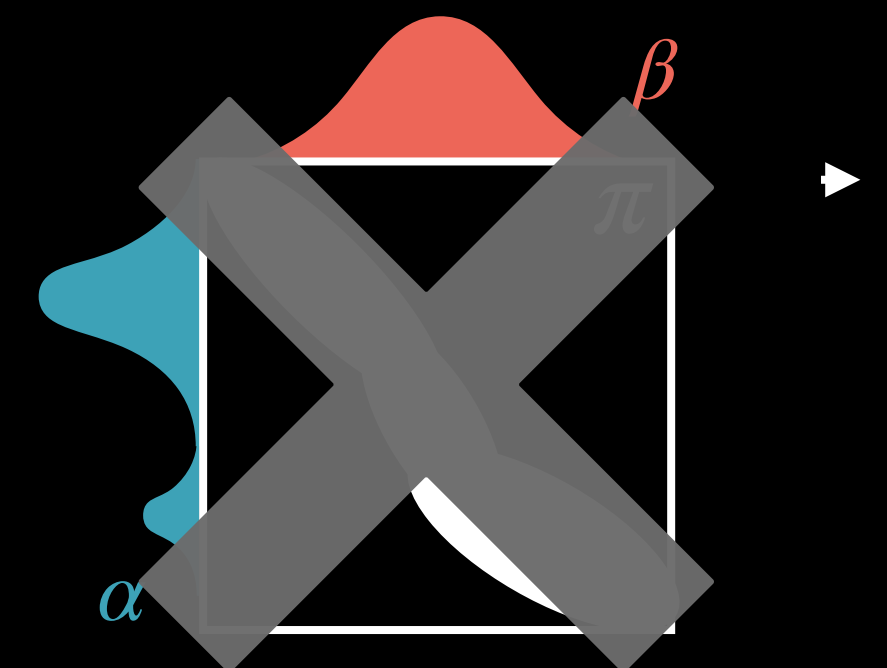
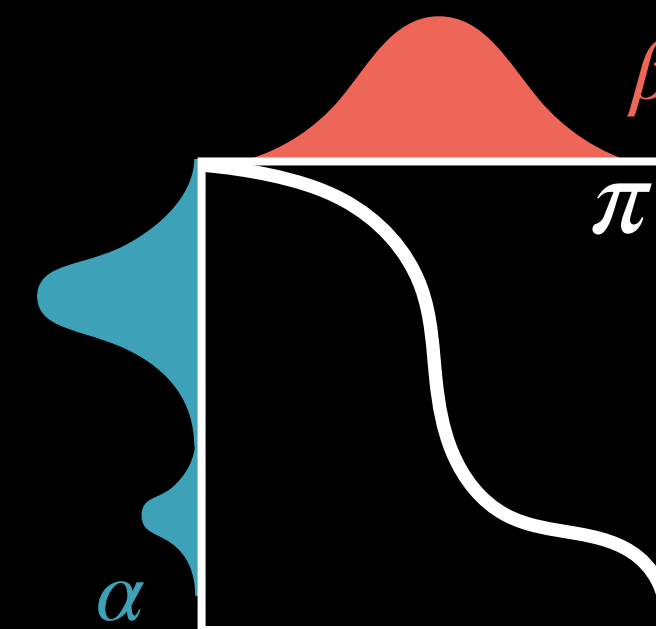
Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) d\pi(x, y)$$

Theorem [Brenier, 1991]:

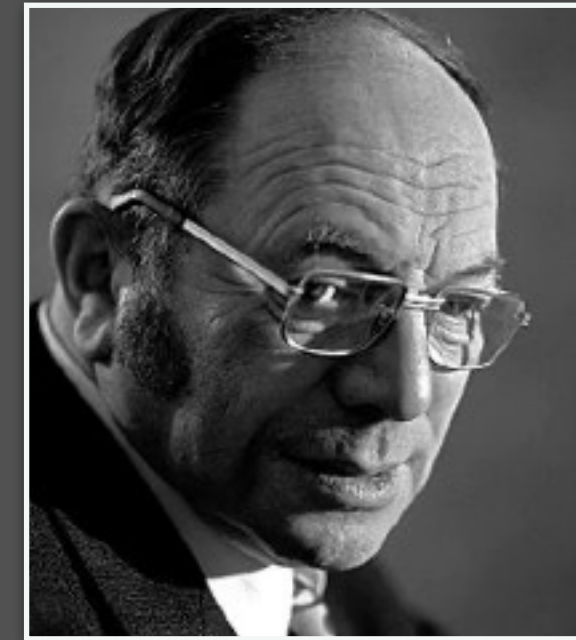
1. There exists a **unique** coupling π that solves the Kantorovich problem
2. The coupling is deterministic, supported on the graph of a Monge map: $\pi = (\text{Id}, T)_\# \alpha$
3. T is the gradient of a convex function
Formally: the unique $T = \nabla \varphi$ such that φ is **convex** and $(\nabla \varphi)_\# \alpha = \beta$



Brenier's Theorem

Assumptions:

1. $c(x, y) = \|x - y\|^2$
2. α has a density $\rho_\alpha = \frac{d\alpha}{dx}$



Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) d\pi(x, y)$$

Theorem [Brenier, 1991]:

1. There exists a **unique** coupling π that solves the Kantorovich problem
2. The coupling is deterministic, supported on the graph of a Monge map: $\pi = (\text{Id}, T)_\# \alpha$
3. T is the gradient of a convex function

Formally: the unique $T = \nabla \varphi$ such that φ is **convex** and $(\nabla \varphi)_\# \alpha = \beta$